

Memory-Centric Agentic Inference — cycles 4-6 clone 2

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Memory-Centric Agentic Inference — cycles 4-6 clone 2

Abstract

Cycles 4-6 did not produce a new clone-2 artifact. They confirmed that the existing clone-2 deliverable, `memory-centric-agentic/experiments/cache_durable_risk_measurement_plan.md`, remains stable, validated, and ready for parent M-EXP-1 integration.

The branch objective was to design the semantic-cache and durable-state risk measurement branch for two deferred constants:

- DC-004: semantic-cache correctness, invalidation, recovery, poisoning, and safe reuse risk.
- DC-003: durable object-store replay latency, tail behavior, pointer validity, retention, and replay-failure risk.

Across all three cycles, researchers, workers, and auditors converged on the same decision: clone 2 should not be reopened unless parent integration exposes a concrete contradiction. The remaining issue is not in the clone-2 plan. It is parent-scope registration and ingestion work: the parent measurement CSVs must ingest the clone-2 CDR-* experiment rows and T-* threshold rows, and `promise_check` orphan warnings must be resolved or explicitly documented before full parent M-EXP-1 validation.

Introduction

The root project investigates whether agentic LLM infrastructure should be designed around memory movement, placement, reuse, compression, and lifetime management rather than arithmetic throughput alone. Earlier synthesis work defined architecture options in which memory reuse becomes progressively more central:

- **Option A:** conventional request/model/KV-centric serving.

- **Option B:** a memory-object-aware runtime that tracks reusable objects such as retrieved context, semantic cache entries, prefix cache entries, tool outputs, provenance pointers, and invalidation state.
- **Option C:** a trajectory/DAG-aware memory fabric that treats branch state, verifier state, durable workspace state, replay dependencies, and multi-agent state as first-class scheduling and placement inputs.

Clone 2 addresses a specific risk in that architecture decision. Semantic caches and prefix caches can reduce recomputation, but raw hit rate is not enough evidence for Option B because hits can be stale, poisoned, tenant-invalid, false-positive, or too expensive to validate and recover. Public prefix-cache and semantic-cache mechanisms provide background for why reuse can matter, but clone 2 keeps correctness and invalidation cost as a measured deferred constant rather than assuming a benefit [12], [13].

Durable state can also reduce recomputation by preserving workspaces, checkpoints, trajectory logs, verifier state, and tool outputs. For Option C, however, durable replay is useful only if replay tails, dependency fan-in, pointer reconstruction, consistency waits, and retention rules do not erase retained value. Storage references such as NVMe specifications provide context for durable tiers, but the relevant agentic quantity is the p50/p95/p99 replay distribution on dependency-critical paths, not generic storage capability [6].

Methodology

This report consolidates clone-2 cycles 4-6. It reports completed work and accepted audit outcomes; it does not re-audit the artifact.

The source material used for this report was:

Cycle	Role	Session ID	Date	What it contains
4	researcher	627afafc-b87d-40e0-9f29-9d0acdebc724	2026-05-11	States clone 2 is complete and stable; directs parent M-EXP-1 ingestion.
4	worker	5a80dec7-6ab4-4a5b-bf3f-12bd2fa6e82c	2026-05-11	Performs no edits; confirms clone 2 is stable input to parent integration.
4	auditor	da403b2f-f8a0-4fce-90f6-0e2524efa436	2026-05-11	Validates the 308-line artifact; records parent-scope orphan warnings.

Cycle	Role	Session ID	Date	What it contains
5	researcher	7ddc59f7-954d-4736-af18-301c7939c048	2026-05-11	Repeats that clone 2 should not be reopened; identifies CSV ingestion as next work.
5	worker	86c209c0-8de5-4519-8066-12578818bc2c	2026-05-11	Performs no edits; restates required experiments, thresholds, and boundaries.
5	auditor	a796fdaa-9e26-4160-9c8a-f77bd4d10259	2026-05-11	Validates artifact content; keeps only parent/root artifact registration as warning.
6	researcher	d6a27062-9da0-4343-86d5-147f8e8f6af6	2026-05-11	Defines parent integration sufficiency criteria for clone-2 coverage.
6	worker	4ce98b80-0dd8-4892-b579-4102f4761967	2026-05-11	Performs no edits; confirms branch is exhausted as a standalone clone.
6	auditor	ff3820ab-d9ab-4431-8207-c6c23adf1f04	2026-05-11	Final supplied validation: clone 2 sufficient; parent integration incomplete.

The reporting pass also used the current artifact, REFERENCES.md, MANIFEST.md, the prior cycles 1-3 clone-2 report, and current validation command outputs.

Results

Cycle 4: Stable Handoff to Parent Integration

Cycle 4 established that clone 2 had moved from design work into handoff status. The researcher session 627afafc-b87d-40e0-9f29-9d0acdebc724 stated that the validated artifact remained memory-centric-agentic/experiments/cache_durable_risk_measurement_plan.md at 308 lines and that no further clone-2 design work was needed.

The worker session 5a80dec7-6ab4-4a5b-bf3f-12bd2fa6e82c made no file changes and ran no commands. This was reported as intentional because the branch had already been validated. The worker preserved the required handoff set:

- 11 CDR experiments: CDR-SEM-001 through CDR-SEM-006, and CDR-DUR-001 through CDR-DUR-005.
- 10 downgrade thresholds: T-SEM-VALID-001 through T-SEM-RECOVERY-005, and T-DUR-TAIL-001 through T-DUR-RETENTION-005.
- DC-004 as semantic-cache correctness, stale/poisoned hit, invalidation, recovery, tenant/cache-salt, and valid-hit retained-value risk.
- DC-003 as durable object-store p50/p95/p99 replay-tail, dependency fan-in, pointer validity, retention, and replay-failure risk.

The auditor session da403b2f-f8a0-4fce-90f6-0e2524efa436 validated the artifact and recorded no clone-scoped defect. It identified one moderate parent-scope warning: promise_check exits successfully but reports orphan artifacts, including parent measurement CSVs and sibling/report artifacts. The audit decision was **VALIDATED**.

Cycle 5: No Reopen, Preserve Mechanism Boundary

Cycle 5 repeated the same branch state with sharper integration guidance. The researcher session 7ddc59f7-954d-4736-af18-301c7939c048 said the next useful work was parent M-EXP-1 CSV ingestion plus ledger/orphan-warning handling. It emphasized that parent ingestion must preserve the mechanism distinction:

- Option B evidence is safe valid semantic retained value, not raw semantic hit rate.
- Option C evidence is tail-safe durable replay value, not median durable latency.

The worker session 86c209c0-8de5-4519-8066-12578818bc2c again made no changes and ran no commands. It concluded that clone 2 was exhausted as a standalone branch and should be consumed by the parent harness.

The auditor session a796fdaa-9e26-4160-9c8a-f77bd4d10259 validated the artifact again. It confirmed that the plan still preserved:

- all 11 experiment IDs,
- all 10 threshold IDs,
- per-experiment instrumentation_fields,
- required safety fields such as tenant_scope, cache_salt, pointer_valid, and retention_hold_state,
- p50/p95/p99 durable replay-tail measurement,
- Option B and Option C downgrade semantics.

The only warning remained parent/root artifact registration. The audit again instructed not to rely on the absent ledger event 523f4e76-04d4-4002-8000-000000000002.

Cycle 6: Final Stability Confirmation

Cycle 6 treated clone 2 as converged. The researcher session d6a27062-9da0-4343-86d5-147f8e8f6af6 stated that the active sub-topic had moved to parent M-EXP-1 integration. It listed the parent tables that must ingest clone-2 rows:

- data/measurement_experiment_specs.csv,
- data/measurement_required_fields.csv,
- data/measurement_thresholds.csv,
- data/measurement_claim_update_matrix.csv,
- data/measurement_synthetic_probe_results.csv.

The worker session 4ce98b80-0dd8-4892-b579-4102f4761967 made no edits. It restated the row identities and concluded that remaining work was parent-scope: CSV ingestion, ledger registration, orphan-warning resolution or documentation, and an integrated M-EXP-1 audit.

The auditor session ff3820ab-d9ab-4431-8207-c6c23adf1f04, also supplied directly as the audit report input, validated clone 2. It reported:

- critical findings: none,
- moderate findings: parent-scope orphan artifact warnings remain,
- minor findings: none,
- artifact line count: 308,
- org_check: green,
- decision: **VALIDATED**.

The audit concluded that reopening or editing cache_durable_risk_measurement_plan.md would not address the remaining issue class. The required next step is parent integration.

Measurement Design Status

The measurement design itself did not change during cycles 4-6. The stable content is summarized here because it is the object being handed off.

DC-004: Semantic-Cache Risk

The DC-004 design treats semantic-cache benefit as a valid-hit retained-value problem, not a raw-hit-rate problem. A semantic-cache hit only counts as useful if it survives correctness, freshness, provenance, tenant/cache-salt, poisoning, validation, recovery, and wrong-reuse gates.

The plan's safe semantic retained-value expression is:

```
V_semantic_safe =  
  P_hit * P_valid_given_hit * C_recompute_avoided  
  - C_lookup  
  - C_validation  
  - C_invalidation  
  - C_recovery  
  - E[C_wrong_reuse]
```

Option B remains justified for retrieval-augmented generation only if this value remains positive after runtime metadata and policy overheads. Unsafe reuse is blocked or downgraded, not merely assigned a lower score.

The six semantic experiments are:

Experiment ID	Measurement
CDR-SEM-001	Raw hit rate versus valid-hit rate.
CDR-SEM-002	Semantic false-positive rate by match-score bucket.
CDR-SEM-003	Stale-hit rate after <code>source_version</code> and invalidation changes.
CDR-SEM-004	Safe hit rate after <code>tenant_scope</code> and <code>cache_salt</code> enforcement.
CDR-SEM-005	Recovery latency and action after invalid cache hit.
CDR-SEM-006	Poisoning or untrusted-provenance rejection rate.

The semantic downgrade thresholds are:

Threshold ID	Decision
T-SEM-VALID-001	Disable semantic-cache retained-value credit when valid-hit value cannot pay lookup, validation, recovery, and wrong-reuse costs.
T-SEM-STALE-002	Require exact provenance or revalidation when stale-hit loss erases recompute savings.
T-SEM-ISOLATION-003	Downgrade Option B if safe hit rate collapses after tenant/cache-salt gates.
T-SEM-POISON-004	Forbid correctness-sensitive reuse when untrusted reuse rate is nonzero without trusted provenance.
T-SEM-RECOVERY-005	Treat lookup as overhead when lookup, validation, and recovery exceed exact retrieval or recompute.

DC-003: Durable-State Replay Risk

The DC-003 design treats durable-state benefit as a replay-tail and dependency-path problem. Durable state supports Option C only when p50/p95/p99 replay latency, consistency wait, fan-in amplification, pointer reconstruction, retention validity, and replay-failure risk remain below retained-value margins.

The plan's durable replay expression is:

```
V_durable_replay =
  P_replay * C_recompute_avoided
  - L_store_read(p)
  - L_consistency(p)
  - C_pointer_validation
  - C_reconstruction
  - E[C_replay_failure]
```

The percentile p must include p50, p95, and p99 because long agentic workflows may block on the slowest object in a replay dependency path.

The five durable experiments are:

Experiment ID	Measurement
CDR-DUR-001	Durable workspace write/read p50/p95/p99 by size and consistency mode.
CDR-DUR-002	Dependency-path max replay latency.
CDR-DUR-003	Verifier/tool-output summary-pointer recovery latency.
CDR-DUR-004	Retention-horizon and hold-state violations.
CDR-DUR-005	Replay success rate under object-store tail injection.

The durable downgrade thresholds are:

Threshold ID	Decision
T-DUR-TAIL-001	Avoid remote durable replay when p95 replay latency exceeds saved recomputation time.
T-DUR-P99-002	Downgrade Option C when max-path p99 exceeds verifier or merge deadline.
T-DUR-FANIN-003	Downgrade when fan-in tail amplification erases retained-value margin.
T-DUR-RECOVERY-004	Disable summary-plus-pointer when pointer is invalid or reconstruction is slower than recompute.
T-DUR-RETENTION-005	Forbid reuse when durability horizon has expired and no retention hold is active.

Integration Status

Clone 2 is complete as a branch artifact but not yet integrated into the parent M-EXP-1 measurement harness.

The parent harness must ingest clone-2 rows into:

Parent table	Required clone-2 contribution
measurement_experiment_specs.csv	Add CDR-SEM-001 through CDR-SEM-006 and CDR-DUR-001 through CDR-DUR-005, each with nonempty per-experiment instrumentation_fields.
measurement_required_fields.csv	Add semantic-cache and durable-state fields such as cache_key_hash, tenant_scope, cache_salt, correctness_label, durable_object_id, dependency_width, read_latency_ms, pointer_valid, and retention_hold_state.
measurement_thresholds.csv	Add all T-SEM-* and T-DUR-* downgrade thresholds.

Parent table	Required clone-2 contribution
measurement_claim_update_matrix.csv	Preserve updates to claims including CL-002, CL-003, CL-006, CL-008, CL-009, and CL-010.
measurement_synthetic_probe_results.csv	Add synthetic probe rows for false positives, stale hits, recovery cost, durable tails, fan-in, pointer validity, and retention expiry.

The current parent measurement CSVs in the workspace are populated for DC-005 trajectory reuse. Clone-2 CDR-* rows remain parent integration input.

The parent integration must preserve deferred-constant separation:

- DC-003: durable replay p95/p99, fan-in, pointer, consistency, retention, and replay-failure risk.
- DC-004: semantic-cache correctness, stale/poisoned hits, invalidation, recovery, isolation, provenance, and valid-hit retained value.
- DC-005: trajectory reuse distribution, a sibling scope.
- DC-006: provenance-validation overhead, a sibling scope.

Discussion

Cycles 4-6 are important because they show convergence rather than construction. The branch stopped producing edits because the artifact already satisfied the scoped objective. Each worker no-op was accepted by audit as the correct behavior: further clone-local edits would not resolve the only remaining issue, which is parent-level registration and ingestion.

The repeated audit result also clarifies the architecture signal. Clone 2 does not say that semantic caches or durable replay are always good. It defines the measurements that decide when they stop being good enough.

For Option B, semantic reuse is credited only when it remains safe and positive after validation. A high raw semantic hit rate can still force downgrade if valid-hit rate collapses, if stale or poisoned entries reach correctness-sensitive paths, if tenant/cache-salt enforcement removes most reusable entries, or if recovery cost exceeds exact retrieval.

For Option C, durable replay is credited only when replay paths remain tail-safe. Median durable latency is not enough. Wide dependency paths, high p99 reads, consistency waits, invalid pointers, slow reconstruction, expired retention, or replay failure can all erase durable retained value and force fallback to local retention, Option B object runtime, recompute, or forbidden reuse.

The main unresolved work is operational: parent M-EXP-1 must encode this branch into the CSV harness and ledger state without flattening the mechanisms into generic cache-hit or storage-latency metrics.

Conclusions and Recommendations

1. Clone 2 is validated and complete for cycles 4-6. The required artifact remains memory-centric-agentic/experiments/cache_durable_risk_measurement_plan.md.
2. No new clone-2 build work occurred in cycles 4-6. That was intentional and accepted by audit because the branch had converged.
3. Option B should only receive semantic-cache reuse credit when safe valid-hit retained

value remains positive after correctness, freshness, provenance, tenant/cache-salt, poisoning, validation, recovery, and wrong-reuse costs.

4. Option C should only receive durable replay credit when p50/p95/p99 tails, dependency fan-in, consistency wait, pointer validity, reconstruction latency, retention state, and replay-failure risk remain below retained-value margins.
5. Unsafe reuse remains a hard downgrade or forbid condition. It is not merely a lower-quality successful reuse case.
6. Parent M-EXP-1 should ingest all clone-2 experiments, thresholds, fields, claim updates, and synthetic probes into the measurement CSVs.
7. Parent validation should not rely on the absent ledger event 523f4e76-04d4-4002-8000-000000000002.
8. Full parent M-EXP-1 validation should wait until `promise_check` orphan warnings are resolved or explicitly documented with ledger evidence.

References

- [6] NVM Express, “Specifications,” NVM Express, accessed 2026-05-11. <https://nvmexpress.org/specifications/>
- [12] vLLM Project, “Automatic Prefix Caching,” vLLM Documentation, accessed 2026-05-11. https://docs.vllm.ai/en/latest/design/prefix_caching/
- [13] Sajal Regmi and Chetan Phakami Pun, “GPT Semantic Cache: Reducing LLM Costs and Latency via Semantic Embedding Caching,” arXiv, 2024. <https://arxiv.org/abs/2411.05276>

Appendix: Implementation Details

Code and Artifact Organization

The primary clone-2 artifact is:

File	Lines	Purpose
memory-centric-agentive/experiments/cache_durable_risk_measurement_plan.md	308	Validated measurement design for DC-004 semantic-cache correctness/invalidation risk and DC-003 durable replay-tail risk.

Sibling experiment-plan artifacts in the workspace are:

File	Lines	Purpose
memory-centric-agentive/experiments/trajectory_reuse_measurement_plan.md	298	DC-005 trajectory-reuse measurement design.
memory-centric-agentive/experiments/provenance_overhead_measurement_plan.md	344	DC-006 provenance-validation overhead measurement design.

The parent measurement CSVs currently observed in the workspace are:

File	Lines	Current status
data/measurement_experiment_specs.csv	8	Populated for DC-005; clone-2 CDR-* rows remain integration input.
data/measurement_required_fields.csv	14	Populated for DC-005; clone-2 fields remain integration input.
data/measurement_thresholds.csv	5	Populated for DC-005; clone-2 τ -* thresholds remain integration input.
data/measurement_claim_update_matrix.csv	12	Populated for DC-005; clone-2 claim updates remain integration input.
data/measurement_synthetic_probe_results.csv	7	Populated for DC-005; clone-2 probe rows remain integration input.

No new figures were produced by clone 2 during cycles 4-6. Existing workspace figures are not specific to this stability pass, so none are embedded in this report.

Workspace Snapshot

MANIFEST.md was updated as a current snapshot after the clone-2 stability pass. The snapshot records:

Category	Count
Python scripts	23
Wolfram scripts	4
Total script lines	7,108
Markdown model/synthesis files	16
Experiment-plan markdown files	3
CSV data/model files	65
Figures	29
Sub-topics completed, assessed, or designed	15

The manifest cross-reference map records that memory-centric-agentic/experiments/cache_durable_risk_measurement_plan.md should feed parent M-EXP-1 measurement CSVs with CDR-SEM-*, CDR-DUR-*, T-SEM-*, and T-DUR-* rows while resolving or documenting parent CSV orphan warnings.

Validation Commands

The reporter pass ran the current workspace checks after updating MANIFEST.md.

org_check passed:

OK: org_check green.

promise_check exited successfully and reported 62 events and 14 plan milestones. It also reported orphan warnings for managed artifacts, including:

- data/dc005_merge_verification_results.csv,
- data/measurement_claim_update_matrix.csv,
- data/measurement_experiment_specs.csv,
- data/measurement_required_fields.csv,
- data/measurement_synthetic_probe_results.csv,
- data/measurement_thresholds.csv,
- reports/cycles/report_cycles_1-3_clone_0.md,
- reports/cycles/report_cycles_1-3_clone_0.pdf,
- reports/cycles/report_cycles_1-3_clone_2.md,
- reports/cycles/report_cycles_1-3_clone_2.pdf,
- tests/verify_dc005_merge_ready.py.

These warnings match the cycle 4-6 audit record: they are parent/root registration issues, not clone-2 artifact defects.

Session Cross-Reference Map

Report section	Source sessions
Abstract and Conclusions	da403b2f-f8a0-4fce-90f6-0e2524efa436, a796fdaa-9e26-4160-9c8a-f77bd4d10259, ff3820ab-d9ab-4431-8207-c6c23adf1f04
Cycle 4 results	627afafc-b87d-40e0-9f29-9d0acdebc724, 5a80dec7-6ab4-4a5b-bf3f-12bd2fa6e82c, da403b2f-f8a0-4fce-90f6-0e2524efa436
Cycle 5 results	7ddc59f7-954d-4736-af18-301c7939c048, 86c209c0-8de5-4519-8066-12578818bc2c, a796fdaa-9e26-4160-9c8a-f77bd4d10259
Cycle 6 results	d6a27062-9da0-4343-86d5-147fbe8f6af6, 4ce98b80-0dd8-4892-b579-4102f4761967, ff3820ab-d9ab-4431-8207-c6c23adf1f04
Measurement design status	memory-centric-agentic/experiments/cache_ durable_risk_measurement_plan.md; all cycle 4-6 audit sessions
Integration status	all cycle 4-6 researcher sessions; all cycle 4-6 auditor sessions
Implementation details	current workspace files, MANIFEST.md, REFERENCES.md, org_check, promise_check

Record Gaps

No clone-2 build records are missing for cycles 4-6. The absence of new artifact edits is itself recorded in all three worker sessions.

The only open gap is parent-scope: the parent M-EXP-1 CSV ingestion and ledger/orphan-warning cleanup were repeatedly identified as next work, but they were not performed by clone 2 during cycles 4-6.